

Fashion

Assignment

For **Spring 2026**, the **Fashion** lab will be used as **Lab 06**.

Before submission, you should be sure to review the [Lab Policy](#) page.

- [PrairieLearn: Lab Model](#)

There is no report component for Lab 06.

- The **model** portion of the lab is due on Saturday, **May 09**.

Background

No background for this lab!

Scenario and Goal

No scenario for this lab!

PyTorch

Unlike other labs, instead of `sklearn`, you will be required to use [PyTorch](#).

To install PyTorch and a necessary data package, follow the instructions in the Computing Policy to [update your virtual environment](#).

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This lab will mostly follow the PyTorch [Quickstart](#) guide.

Compute

Completing this lab necessitates access to powerful computational resources.

```
import torch
from torchvision import datasets
from torchvision.transforms import ToTensor

if torch.cuda.is_available():
    device = "cuda"
elif torch.backends.mps.is_available():
    device = "mps"
else:
    device = "cpu"

print(f"Using {device} device!")
```

If your machine does not allow for `mps` or `cuda` consider using:

- [Illinois Computes Research Notebooks](#)
 - Utilize the CS 307 [Remote Setup](#) to get started.
 - Be sure to select the PyTorch option to utilize a GPU.

Data

The data (and task) for this lab originally comes from Zalando Research.

- [GitHub: Fashion-MNIST](#)

We will access this data through tools available in `pytorch`.

- [PyTorch: Fashion-MNIST](#)

Data in Python

The following code can be used to download or re-access the data.

```
# download training data from open datasets
training_data = datasets.FashionMNIST(
```

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```
    root="data",
    train=True,
    download=True,
    transform=ToTensor(),
)

# download test data from open datasets
test_data = datasets.FashionMNIST(
    root="data",
    train=False,
    download=True,
    transform=ToTensor(),
)
```

The first time this code runs, it will create a directory named data (in the same directory of whatever notebook you are working in), place the data there, and load it. After the first time, running this code again will simply read in the downloaded data.

Like MNIST, there are ten classes for the `y` data, also represented by integers. The following maps, in order, to the integers 0 to 9.

```
classes = [
    "T-shirt/top",
    "Trouser",
    "Pullover",
    "Dress",
    "Coat",
    "Sandal",
    "Shirt",
    "Sneaker",
    "Bag",
    "Ankle boot",
]
```

The `x` data are 28x28 greyscale images of articles of clothing.



Sample Statistics

Before modeling, be sure to look at the data. Calculate the summary statistics requested on PrairieLearn.

Models

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For this lab you will select one model to submit to the autograder. You may use any modeling techniques you'd like, so long as it meets these requirements:

- Models must be built in PyTorch using `nn.Module` class.
- Your model should be created with `torch` version 2.9.1 or newer.
- Your model should be created with `torchvision` version 0.24.1 or newer.
- Your model should be serialized to TorchScript.
- Your serialized model must be less than 5MB.

To obtain the maximum points via the autograder, your model must outperform the following metrics:

- Test Accuracy: 0.9

Model Persistence

To submit your model to the autograder, you will first need to serialize your model. With a `pytorch` model named `model`, convert it to TorchScript. Then, write this object to disk.

```
# convert to TorchScript
model_scripted = torch.jit.script(model)

# write to disk
model_scripted.save("fashion.pt")
```

This website is developed and maintained by [David Dalpiaz](#) for [CS 307](#) at [UIUC](#).

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